Day 38 EBIKE101 Morning Quiz — Student Copy

Program: CTS Cohort 6

Course: EBIKE101

Date: Tuesday, 2026-09-08

Topic: E-bike Homunculous diagnosis and finish-work planning

Name: _____

Directions

Circle the best answer for each question. This quiz is about safe diagnosis, not guessing bravely at wires like a tiny electrician raccoon.

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1. Before diagnosing an e-bike electrical problem, what should happen first?

A. Connect every loose wire and see what turns on

B. Confirm the system is safe to inspect and de-energized when appropriate

C. Spin the motor by hand as fast as possible

D. Replace the controller immediately

2. What is the best first step when looking at an unfinished e-bike build?

A. Make a full system inventory and note what is present, missing, loose, or unknown

B. Assume the last person wired it correctly

C. Start cutting zip ties so every wire is free

D. Connect the battery and test under load

3. What is the job of a fuse or breaker in an e-bike power circuit?

A. Make the motor run faster

B. Stop unsafe current before wiring or parts are damaged

C. Increase battery voltage

D. Turn pedal assist on automatically

4. If the display does not power on, which item is a reasonable early check?

A. Tire tread pattern

B. Battery voltage, main switch, fuse, and display/controller connector

C. Seat height only

D. Rear reflector angle

5. What does a missing throttle usually mean during diagnosis?

A. The whole project is impossible

B. The team should document the missing control input and verify whether the system can operate by PAS/display without it

C. The battery must be bad

D. The motor wires should be reversed

6. Why should wire color not be trusted by itself?

A. Wire color is always random

B. Wire color is a clue, but repairs, adapters, and different manufacturers can change meanings

C. Only red wires carry electricity

D. Black wires are never connected to anything

7. What is the safest way to use a continuity test?

A. On a de-energized circuit or component, following meter instructions

B. Across a live battery pack

C. Across the controller's main power leads while the battery is connected

D. Anywhere, because continuity mode is harmless

8. What can happen if an e-bike brake cutoff sensor is active or misadjusted?

A. The controller may prevent motor output

B. The battery voltage will double

C. The tire pressure will drop

D. The chain will automatically tighten

9. What does PAS stand for in most e-bike systems?

A. Pedal Assist Sensor/System

B. Power Amplifier Switch

C. Positive Axle Support

D. Panel Alignment Screw

10. Why does motor cable strain relief matter?

A. It makes the bike look expensive

B. It prevents cable damage from pulling, rubbing, bending, or wheel movement

C. It increases top speed

D. It changes the tire size

11. If a fuse is blown, what should the team do before simply replacing it?

A. Bypass it with wire

B. Ask what caused the fuse to blow and inspect for shorts, wrong connections, or overload

C. Install the largest fuse available

D. Tape over the fuse holder

12. What does it mean to trace a circuit?

A. Draw a pretty version of the bike after it works

B. Follow the actual path from source, through protection/control, to the load

C. Guess which wire is correct based on color

D. Skip testing because the labels look professional

13. Which mechanical item is critical before powered wheel testing?

A. Handlebar grip color

B. Axle nuts/torque arms, wheel security, brake function, and clear chain/wheel movement

C. Sticker placement

D. Bell volume

14. What is a good practice for first powered motor testing in class?

A. Full throttle with someone holding the rear wheel

B. Secure the bike, keep people clear of moving parts, use low power, and have instructor control

C. Ride it through the shop immediately

D. Test with no brakes installed

15. What should be recorded in a diagnostic work log?

A. Only the final answer

B. Symptoms, observations, measurements, changes made, parts needed, and next step

C. Only who touched the bike last

D. Nothing if the team feels confident

16. What does a controller usually do in an e-bike system?

A. Converts rider/display inputs into controlled power sent to the motor

B. Inflates the tires

C. Replaces the chain

D. Makes the battery physically larger

17. What should happen if a component smells burnt, gets unusually hot, or shows melted wiring?

A. Keep testing to gather more data

B. Stop, disconnect power if safe, and get instructor review

C. Spray it with water

D. Wrap it in tape and continue

18. Why is connector inspection important?

A. Small pins, bent terminals, loose plugs, corrosion, or mismatched connectors can stop a system from working

B. Connectors are only cosmetic

C. Connectors never fail

D. Every connector can plug into every other connector safely

19. What is the difference between a symptom and a root cause?

A. They are always the same thing

B. A symptom is what you observe; a root cause is the underlying reason it is happening

C. A root cause is always the most expensive part

D. A symptom only matters if the motor spins

20. When is the Homunculous closer to “finished”?

A. When one person says “looks good”

B. When mechanical safety, electrical safety, control function, wiring support, needed parts, and documentation are verified

C. When every wire is hidden

D. When the fastest student rides it first

Exit check

Write one thing your team must verify before this e-bike should be powered or ridden:

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